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1. A laboratory mixing vessel initially contains 4 litres of solution in which 4 mg of dye is dissolved.

A wash solution enters the vessel at a constant rate of 300 ml per minute.

At the same time, well-mixed liquid leaves the vessel at a constant rate of 200 ml per minute.

The concentration of dye in the liquid entering the vessel, at time t minutes after pumping begins, is modelled by $2e^{-0.1t}$ mg per litre.

Let m be the amount of dye, in milligrams, in the vessel at time t minutes.

- (a) Show that the volume of liquid in the vessel is $(4 + 0.1t)$ litres, and hence show that m satisfies

$$\frac{dm}{dt} = 0.6e^{-0.1t} - \frac{2m}{40 + t} \quad [4]$$

- (b) Solve this differential equation, given that $m = 4$ when $t = 0$, to find an expression for m in terms of t [8]

- (c) After 20 minutes, the concentration of dye in the vessel was measured as 0.82 mg per litre. Assess the model in light of this information, giving a reason for your answer [2]

2. A particle P , of mass 2 kg, moves in a straight line and has velocity $v \text{ m s}^{-1}$.

At time t seconds, where $t \geq 0$, a variable force of $\frac{12t - 4v}{1 + t}$ N acts on P .

There are no other forces acting on P . Initially, when $t = 0$, P has velocity -5 m s^{-1} .

The motion of P can be modelled by the differential equation

$$\frac{dv}{dt} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t .

(a) Find the functions $P(t)$ and $Q(t)$ [2]

(b) Using an integrating factor, determine the first time at which P is stationary according to the model [8]

3. A brief flushing cycle in an industrial pipe is to be modelled. The flow rate of liquid, R litres per second, at time t seconds is measured from the instant the valve is opened. The cycle starts when $t = 0$ and there is initially no flow through the pipe. When $t = 1$, the flow rate is measured as $\pi e^{-1} \sin(1)$ litres per second.

While the cycle is occurring, R is modelled by the differential equation

$$\frac{dR}{dt} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

- (a) By using an integrating factor show that, according to the model, while the cycle is occurring,

$$R = e^{-t} \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right) \quad [6]$$

The cycle ends when there is again no flow through the pipe.

- (b) Determine the length of time that the cycle lasts according to the model [2]
- (c) Determine, according to the model, the rate of increase of the flow rate at the start of the cycle. Give your answer in exact form [3]

4. A mixing tank initially contains 600 litres of pure water.

Brine flows into the tank at a constant rate of 20 litres per hour. Each litre of brine contains 3 grams of dissolved salt.

The well-stirred mixture is pumped out of the tank at a constant rate of 30 litres per hour.

It is assumed that the salt is instantly and uniformly mixed throughout the liquid.

Given that there are x grams of salt in the tank after t hours,

(a) show that, while the tank still contains liquid, the situation can be modelled by

$$\frac{dx}{dt} = 60 - \frac{3x}{60 - t} \quad [4]$$

(b) Hence find the number of grams of salt in the tank after 30 hours [5]

(c) State one limitation of this model and explain briefly how the model could be refined [1]

5. The concentration C , in suitable units, of a pollutant in a reservoir at time t hours after a filtration system is switched on is to be modelled by a differential equation. Initially, the concentration is A , and its long-term value is 0.

- (a) One simple model is to assume that the rate of decrease of concentration is directly proportional to C .

(i) Formulate a differential equation for this model [1]

(ii) Verify that $C = Ae^{-kt}$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition [3]

An alternative model uses the differential equation

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = R(t)$$

where $R(t)$ is a function of t .

- (b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$1 + t^2 \quad [3]$$

- (c) Suppose that $R(t) = 0$.

(i) Show that

$$C = \frac{A}{1+t^2} \quad [4]$$

(ii) Find the time predicted by this model for the concentration to fall to half its initial value. Give your answer correct to the nearest minute [2]

- (d) Now suppose that

$$R(t) = \frac{te^{-t}}{1+t^2}$$

Show that

$$C = \frac{A + 1 - (1+t)e^{-t}}{1+t^2} \quad [6]$$

It is found that the initial concentration is 5, and the concentration falls to half this value after 63 minutes.

- (e) Determine which of the models proposed in parts (c) and (d) is more consistent with these data [2]

6. A processing tank initially contains 15 litres of water in which 6 mg of dye is dissolved.

A dye solution enters the tank at a constant rate of 200 ml per minute. The concentration of dye in this incoming solution, at time t minutes after the flow begins, is modelled by $(4 + e^{-0.1t})$ mg per litre.

At the same time, pure water enters through a second pipe at a constant rate of 100 ml per minute. The liquid in the tank is continually mixed and leaves the tank at a constant rate of 300 ml per minute.

Let m be the amount of dye, in milligrams, in the tank at time t minutes after the flow begins.

- (a) (i) Write down the volume of liquid in the tank at time t
(ii) Hence show that m satisfies the differential equation

$$\frac{dm}{dt} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50} \quad [4]$$

- (b) By solving the differential equation, find an expression for m in terms of t [8]

After 30 minutes, the concentration of dye in the tank was measured as 1.52 mg per litre.

- (c) Assess the model in light of this information, giving a reason for your answer [2]

7. A particle P , of mass 2 kg, moves in a straight line and has velocity $v \text{ m s}^{-1}$.

At time t seconds, where $t \geq 0$, a variable force of $-2t(v + 3) \text{ N}$ acts on P . There are no other forces acting on P . Initially, when $t = 0$, P has velocity 5 m s^{-1} .

The motion of P can be modelled by the differential equation

$$\frac{dv}{dt} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t .

(a) Find the functions $P(t)$ and $Q(t)$ [2]

(b) Using an integrating factor, determine the first time at which P is stationary according to the model [8]

8. A small rescue boat moves in a straight line on calm water. It starts from rest, is driven by its engine for 5 seconds and then, for the next 5 seconds, the engine is cut and a drogue is deployed. The boat comes to rest at the end of the 10 seconds. The total mass of the boat and crew is m kg, and at time t seconds, for $0 \leq t \leq 10$, the boat's velocity is v m s⁻¹.

A resistance to motion, modelled by a force of magnitude $0.2mv$ N, acts on the boat throughout the 10 seconds.

- (a) Explain why modelling the resistance to motion in this way is likely to be more realistic than assuming this force is constant [1]

- (b) During the drogue phase of the motion, for $5 \leq t \leq 10$, the drogue exerts an additional constant resistance force of magnitude m N and the engine provides no driving force. Show that, for $5 \leq t \leq 10$,

$$\frac{dv}{dt} + 0.2v = -1 \quad [1]$$

- (c) (i) Solve the differential equation in part (b) [4]
(ii) Hence find the velocity of the boat when $t = 5$ [1]

- (d) During the powered phase ($0 \leq t \leq 5$), the propeller provides a driving force of magnitude directly proportional to t^2 . Show that, for $0 \leq t \leq 5$,

$$\frac{dv}{dt} + 0.2v = \lambda t^2 \quad [1]$$

where λ is a positive constant

- (e) (i) Show by integration that, for $0 \leq t \leq 5$,

$$v = 5\lambda(t^2 - 10t + 50 - 50e^{-t/5}) \quad [6]$$

- (ii) Hence find λ [2]

- (f) Find the total distance, to the nearest metre, travelled by the boat during the motion [6]

9. A winch raises a lift cage vertically. The cage starts from rest and comes to rest again 8 seconds later. The combined mass of the cage and its load is m kg, and at time t seconds, for $0 \leq t \leq 8$, the cage has upward velocity $v \text{ m s}^{-1}$.

A resistance to the motion, modelled by a force of magnitude $0.25mv$ N, acts on the cage throughout the 8 seconds.

- (a) Explain why modelling the resistance to the motion in this way is likely to be more realistic than assuming this force is constant [1]

- (b) During the final stage of the motion, for $4 \leq t \leq 8$, the winch provides a constant upward tension of magnitude $0.75mg$ N. Show that, for $4 \leq t \leq 8$,

$$\frac{dv}{dt} + 0.25v = -\frac{1}{4}g$$
 [1]

- (c) (i) Solve the differential equation in part (b) [5]

- (ii) Hence find the velocity of the cage when $t = 4$ [1]

- (d) During the first stage of the motion, for $0 \leq t \leq 4$, the winch provides an upward tension of magnitude $m(g + \lambda t)$ N, where λ is a positive constant. Show that, for $0 \leq t \leq 4$,

$$\frac{dv}{dt} + 0.25v = \lambda t$$
 [1]

- (e) (i) Show by integration that, for $0 \leq t \leq 4$,

$$v = 4\lambda(t - 4 + 4e^{-t/4})$$
 [5]

- (ii) Hence find λ [2]

- (f) Find the total distance, to the nearest metre, through which the cage rises during the motion [6]

- 10.** A short filling pulse in a buffer tank is to be modelled. The volume of liquid in the tank at time t seconds is V litres. The pulse starts when $t = 0$, and initially the tank is empty. After 1 second the volume is measured to be 4 litres. While the pulse is occurring, V is modelled by the differential equation

$$(3t - t^2) \frac{dV}{dt} = \frac{(3t - t^2)^2}{1 + (t - 1)^2} + (3 - 2t)V$$

- (a) By using an integrating factor show that, according to the model, while the pulse is occurring,

$$V = (3t - t^2) (\tan^{-1}(t - 1) + 2) \quad [6]$$

The pulse ends when the tank is again empty.

- (b) Determine the length of time that the pulse lasts according to the model [2]
- (c) Determine, according to the model, the rate of increase of the volume at the start of the pulse. Give your answer in an exact form [3]

11. A horticultural mixing tank initially contains 600 litres of pure water.

A fertiliser solution flows into the tank at a constant rate of 6 litres per minute. The incoming solution contains 2 grams of fertiliser in every litre of solution.

At the same time, liquid is pumped out of the well-stirred tank at a constant rate of 9 litres per minute.

It is assumed that the fertiliser instantly dissolves uniformly throughout the tank.

Given that there are x grams of fertiliser in the tank after t minutes,

- (a) Show that, while the tank still contains liquid, the situation can be modelled by the differential equation

$$\frac{dx}{dt} = 12 - \frac{3x}{200 - t} \quad [4]$$

- (b) Hence determine after how many minutes the concentration of fertiliser in the tank first reaches 1.5 grams per litre. [5]

- (c) Explain briefly one way in which the model could be refined. [1]

- 12.** A mixing tank initially contains 2 litres of solution in which 4 grams of dye are dissolved.
A dye solution containing 10 grams of dye per litre is pumped into the tank at a constant rate of r litres per minute and mixes instantly and uniformly with the liquid already in the tank.
At the same time, liquid is removed from the tank at a constant rate of s litres per minute.
After t minutes, the tank contains m grams of dye.

- (a) (i) Show that the concentration of dye in the tank after t minutes is

$$\frac{m}{2 + (r - s)t}$$

grams per litre.

[2]

- (ii) Hence show that

$$\frac{dm}{dt} + \frac{sm}{2 + (r - s)t} = 10r$$

[2]

- (b) First, consider the case where $s = r$.

- (i) Solve the differential equation to find m in terms of r and t . [4]

- (ii) Given that after 2 minutes the concentration of dye in the tank is 8 grams per litre, find r . [2]

- (c) Now consider the case where $s = 2r$.

- (i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{2}{r}$. [1]

- (ii) Find the maximum mass of dye in the tank. [9]

- 13.** The concentration C , in suitable units, of a drug in a patient's bloodstream at time t hours after an intravenous infusion begins is to be modelled. Initially, the concentration is zero, and its long-term value is M .

(a) One simple model assumes that the rate of change of concentration is directly proportional to $M - C$.

(i) Formulate a differential equation for this model. [1]

(ii) Verify that $C = M(1 - e^{-kt})$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition. [3]

An alternative model uses, for $t > 0$, the differential equation

$$\frac{dC}{dt} - \frac{t+2}{t(1+t+t^2)} C = S(t),$$

where $S(t)$ is a function of t .

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$\frac{1+t+t^2}{t^2} \quad [8]$$

(c) Suppose that $S(t) = 0$.

(i) Show that

$$C = \frac{Mt^2}{1+t+t^2} \quad [4]$$

(ii) Find the time predicted by this model for the concentration to reach half its long-term value. Give your answer correct to the nearest minute. [2]

(d) Now suppose that

$$S(t) = \frac{3t^2 e^{-t}}{1+t+t^2}$$

Show that

$$C = \frac{Mt^2 - 3t^2 e^{-t}}{1+t+t^2} \quad [5]$$

(e) It is found that the long-term concentration is 10, and C reaches half this value after 105 minutes. Determine which of the models proposed in parts (c) and (d) is more consistent with this data. [2]

- 14.** A drinks manufacturer uses a mixing tank that initially contains 4 litres of water. A flavouring liquid, which is 60% fruit concentrate by volume, is pumped into the tank at a constant rate of p litres per minute. At the same time, the contents of the tank are thoroughly mixed and drawn off at a constant rate of q litres per minute.

After t minutes, let x litres be the amount of fruit concentrate in the tank. You may assume that mixing is instantaneous and uniform.

- (a)** (i) Show that the proportion of fruit concentrate in the tank after t minutes is

$$\frac{x}{4 + (p - q)t} \quad [2]$$

- (ii) Hence show that

$$\frac{dx}{dt} + \frac{qx}{4 + (p - q)t} = \frac{3p}{5} \quad [2]$$

- (b)** Now consider the case where $q = p$.

- (i) Solve the differential equation to find x in terms of p and t . [4]

- (ii) Given that after 4 minutes the liquid in the tank is 30% fruit concentrate, find the value of p . [2]

- (c)** Now consider the case where $q = 2p$.

- (i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{4}{p}$. [1]

- (ii) Find the maximum amount of fruit concentrate in the tank before the tank empties. [9]